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Inventor(s)	Chae; Jung Yong et al.

Semiconductor device including a through silicon via structure and method of fabricating the same

Abstract

A semiconductor device may include a via hole, a first electrode, a second electrode and a first protecting insulation layer. The via hole may be formed to penetrate a substrate. The first electrode may include an electrode segment formed on a surface of the via hole. The second electrode may be formed on the first electrode along the surface of the via hole. The second electrode may include two ends that are positioned below a surface of the substrate. The first protecting insulation layer may be formed on the second electrode along the surface of the via hole. The first protecting insulation layer may include both ends that upwardly protrude from the both ends of the second electrode.

Inventors:	Chae; Jung Yong (Icheon-si, KR), Cho; Jin Hee (Icheon-si, KR)
Applicant:	SK hynix Inc. (Icheon-si, KR)
Family ID:	1000008778321
Assignee:	SK HYNIX INC. (Icheon-si, KR)
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Primary Examiner: Luu; Chuong A

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATION (1) This patent document is a continuation of U.S. patent application Ser. No. 17/219,336, filed on Mar. 31, 2021, which claims priority under 35 U.S.C. § 119(a) to Korean application number 10-2020-0179951, filed on Dec. 21, 2020, in the Korean Intellectual Property Office, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The technology and implementations disclosed in this patent document generally relate to a semiconductor device, and more particularly, to a semiconductor device with a through silicon via (TSV) structure, and a method of manufacturing the semiconductor device.

BACKGROUND

(2) As a three-dimensional (3D) package including a plurality of semiconductor chips in one semiconductor device is getting more developed, a through silicon via (TSV) technology configured to form an electrical connection vertically penetrating a substrate or a die becomes more important. In order to increase a performance and reliability of the 3D package, providing the TSV structure with stable operations characteristics and high reliability is required.

SUMMARY

(3) Example embodiments of the disclosed technology provide a semiconductor device including an improved TSV structure.

(4) Example embodiments of the disclosed technology also provide a method of manufacturing the above-mentioned semiconductor device.

(5) In example embodiments of the disclosed technology, a semiconductor device may include a via hole, a first electrode, a second electrode and a first protecting insulation layer. The via hole may be formed to penetrate a substrate. The first electrode may include an electrode segment formed on a surface of the via hole. The second electrode may be formed on the first electrode along the surface of the via hole. The second electrode may include two ends that are positioned below a surface of the substrate. The first protecting insulation layer may be formed on the second electrode along the surface of the via hole. The first protecting insulation layer may include both ends that upwardly protrude from the both ends of the second electrode.

(6) In example embodiments, the semiconductor device may further include a second protecting insulation layer and a capping layer. The second protecting insulation layer may be configured to cover the both ends of the second electrode exposed by the first electrode and the first protecting insulation layer. The capping layer may be configured to cover the substrate and fill the via hole.

(7) In example embodiments of the disclosed technology, a semiconductor device may include a stack structure, a via hole, a first electrode, a second electrode and a first protecting insulation layer. The stack structure may include a lower structure and an upper structure. The lower structure may include a lower substrate, a lower insulating interlayer disposed on the lower substrate, and a lower circuit disposed in the lower insulating interlayer. The lower circuit may include a lower via pad. The upper structure may include an upper substrate, an upper insulating interlayer disposed on the upper substrate, and an upper circuit disposed in the upper insulating interlayer. The via hole may be formed at the stack structure to expose the lower via pad. The first electrode may be formed on a surface of the via hole. The second electrode may be formed on the first electrode along the surface of the via hole. The second electrode may include both ends positioned below a surface of the stack structure. The first protecting insulation layer may be formed on the second electrode along the surface of the via hole. The first protecting insulation layer may include both ends upwardly protruding from the both ends of the second electrode.

(8) In example embodiments, the semiconductor device may further include a second protecting insulation layer and a capping layer. The second protecting insulation layer may be configured to cover the both ends of the second electrode exposed by the first electrode and the first protecting insulation layer. The capping layer may be configured to cover the stack structure and fill the via hole.

(9) In example embodiments of the disclosed technology, according to a method of fabricating a semiconductor device, a substrate may be etched to form a via hole. A first electrode may be formed on a surface of the via hole. A conductive layer and an insulation layer may be formed on a surface of the substrate. The conductive layer and the insulation layer may be etched to form a second electrode on the first electrode and a first protecting insulation layer on the second electrode. The second electrode may be formed on the first electrode along the surface of the via hole. The first protecting insulation layer may be formed on the second electrode along the surface of the via hole. The second electrode may include both ends that may be positioned below a surface of the substrate. The first protecting insulation layer may include both ends that upwardly protrude from the both ends of the second electrode.

(10) In example embodiments, the method may further include forming a second protecting insulation layer to cover the both ends of the second electrode exposed by the first electrode and the first protecting insulation layer, and forming a capping layer to cover the substrate and to fill the via hole.

(11) According to example embodiments, the both ends of the second electrode may be positioned lower than the surface of the substrate. Further, the both ends of the first protecting insulation layer may be protruded from the both ends of the second electrode. Thus, an oxidation of the second electrode caused by moisture infiltration may be prevented.

(12) Further, because the both ends of the first protecting insulation layer may be protruded from the both ends of the second electrode, the both ends of the second electrode may have a flat surface parallel to the surface of the substrate. Thus, the second protecting insulation layer configured to cover the second electrode may be readily formed together with the first protecting insulation layer. As a result, the oxidation of the second electrode caused by the moisture infiltration may be effectively prevented.

(13) Furthermore, the first protecting insulation layer and the second protecting insulation layer may include a non-metal based insulation layer. Thus, although the moisture may infiltrate into the second electrode, a galvanic corrosion may not be generated between the second electrode and the first protecting insulation layer, and between the second electrode and the second protecting insulation layer.

(14) As a result, the oxidation and the corrosion of the second electrode caused by the moisture infiltration may be effectively prevented so that the semiconductor device including the TSV structure may have improved reliability.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and another aspects, features and advantages of the subject matter of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings.

(2) FIG. 1A is a cross-sectional view illustrating an example of a semiconductor device in accordance with an implementation of the disclosed technology.

(3) FIG. 1B is an enlarged view illustrating a portion “A” in FIG. 1A.

(4) FIGS. 2A to 2E are cross-sectional views illustrating an example of a method of fabricating a semiconductor device in accordance with an implementation of the disclosed technology.

DETAILED DESCRIPTION

(5) Various embodiments of the disclosed technology will be described in greater detail with reference to the accompanying drawings.

(6) Hereinafter, example embodiments may provide a semiconductor device including an electrically conductive path through a substrate such as a through-silicon vias (TSV) structure as an electrical connection to provide improved reliability and a method of fabricating the semiconductor device. The TSV structure having the improved reliability may be capable of preventing an oxidation caused a moisture infiltration. When an electrode of the TSV structure may be partially protruded from a via hole, which can be caused by a process, a moisture may easily infiltrate into the protruded electrode. Although a capping layer may cover the TSV structure, such protrusion of the electrode of the TSV structure may decrease the reliability of the TSV structure. Thus, example embodiments may provide the semiconductor device including the TSV structure and the method of fabricating the semiconductor device that may be capable of preventing the above-mentioned problem.

(7) FIG. 1A is a cross-sectional view illustrating a semiconductor device in accordance with example embodiments, and FIG. 1B is an enlarged view illustrating a portion “A” in FIG. 1A.

(8) Referring to FIGS. 1A and 1B, a semiconductor device **10** of example embodiments may include a stack structure **20**. The stack structure **20** may include a lower structure **100**, an upper structure **200** stacked on the lower structure **100** and a TSV structure **300** electrically connected between the lower structure **100** and the upper structure **200**. The stack structure **20** may include a through electrode area (TEA) in which the TSV structure **300** may be positioned, and a pad area (PA) in which a pad structure **400** may be positioned.

(9) The lower structure **100** in the stack structure **20** may include a lower substrate **110** and a lower circuit **140**. The lower circuit **140** may be arranged in a lower insulating interlayer **150** on the lower substrate **110**. The upper structure **200** may include an upper substrate **210** and an upper circuit **240**. The upper circuit **240** may be arranged in an upper insulating interlayer **250** over or on the upper substrate **210**. The upper structure **200** may be stacked over or on the lower structure **100** to form the stack structure **20**. In the upper structure **200** and the lower structure **100** of the stack structure **20**, the upper insulating interlayer **250** and the lower insulating interlayer **150** may be bonded to each other by contacting the upper insulating interlayer **250** with the lower insulating interlayer **150**, not restricted within a specific manner. Alternatively, in the upper structure **200** and the lower structure **100** of the stack structure **20**, the lower insulating interlayer **150** may make contact with the upper substrate **210** to bond the lower insulating interlayer **150** to the upper substrate **210**.

(10) The lower substrate **110** and the upper substrate **210** may include at least one of a bulk single crystalline silicon wafer, a silicon-on-insulator (SOI) wafer, a compound semiconductor wafer such as Si—Ge, a wafer including a silicon epitaxial layer, or others.

(11) The lower circuit **140** may include a lower wiring **120** and a lower via pad **130**. The lower via pad **130** may be positioned in the TEA. The lower via pad **130** may electrically make contact with a lower end of the TSV structure **300**. The upper circuit **240** may include an upper wiring **220** and an upper via pad **230**. The upper via pad **230** may be positioned in the TEA. The upper via pad **230** may electrically make contact with the TSV structure **300**, not restricted within a specific connection structure. Alternatively, the upper via pad **230** may be spaced apart from the TSV structure **300** to be electrically isolated from the TSV structure **300**.

(12) The lower wiring **120**, the lower via pad **130**, the upper wiring **220** and the upper via pad **230** may include at least one of a metal such as tungsten, aluminum, copper, etc., a metal silicide such as tungsten silicide, titanium silicide, etc., a metal compound such as tungsten nitride, titanium nitride, etc., a doped silicon, or others.

(13) The TSV structure **300** in the TEA may include a via hole **310**, a first electrode **320**, a second electrode **330**, a first protecting insulation layer **340** and a second protecting insulation layer **350**.

The via hole **310** may be formed through the stack structure **20**. As illustrated in the example in FIG. **1A**, the via hole **310** penetrates through the upper structure **200** to reach into the lower structure **100** (which includes the lower circuit **140**). The first electrode **320** may be formed to include an electrode segment formed inside the via hole **310** on a surface of the via hole **310** and to form an electrical connection which may, as in the example shown in FIG. **1A**, to connect the pad structure **400** in the pad area PA and the lower circuit **140** in the lower structure **100**. The second electrode **330** may be formed inside the via hole **310** and on the first electrode **320** along the surface of the via hole **310**. The first protecting insulation layer **340** may be formed inside the via hole **310** and on the second electrode **330** along the surface of the via hole **310**. The second protecting insulation layer **350** may be configured to cover the two ends of the second electrode **330** that are not covered by the first protecting insulation layer **340**. In the example in FIG. **1A**, both insulation layers **340** and **350** collectively cover the second electrode **330**. The TSV structure **300** may further include a capping layer **560** formed on the stack structure **20** to fill the via hole **310**. The capping layer **560** may include an insulation material.

(14) The via hole **310** may be formed through the upper substrate **210**, the upper via pad **230** and the upper insulating interlayer **250**. The via hole **310** may be partially formed through the lower insulating interlayer **150** to expose the lower via pad **130**. In example embodiments, the via hole **310** may include single hole pattern configured to expose the lower via pad **130**. However, other implementations are possible. For example, the via hole **310** may include a plurality of hole patterns configured to expose the lower via pad **130**.

(15) The first electrode **320** may be formed along the surface of the via hole **310**. The first electrode **320** may have a uniform thickness. Thus, the first electrode **320** having the uniform thickness may be formed on a bottom surface and a side surface of the via hole **310**. Both ends of the first electrode **320** may be extended to the upper substrate **210**. The first electrode **320** may be electrically connected between the TSV structure **300** and the pad structure **400**. The first electrode **320** may include at least one of a metal, a metal silicide or a metal compound. For example, the first electrode **320** may include tungsten.

(16) The second electrode **330** may be formed on the first electrode **320** in the via hole **310** along the surface of the via hole **310**. Thus, the second electrode **330** may have a cylindrical shape. The second electrode **330** may have a uniform thickness. Alternatively, a portion of the second electrode **330** on the bottom surface of the via hole **310** may have a thickness thicker than a thickness of a portion of the second electrode **330** on an upper portion of the side surface of the via hole **310**. In example embodiments, the second electrode **330** having the uniform thickness may be formed along the surface of the via hole **310**.

(17) In order to prevent the oxidation caused by the moisture infiltration, the both ends of the second electrode **330** may be positioned lower than a surface of the stack structure **20**, i.e., the surface **S1** of the upper substrate **210**. Since the first protecting insulation layer **340** has both ends protruding from the both ends of the second electrode **330**, the both ends of the second electrode **330** may have a flat surface **S2** parallel to the surface **S1** of the upper substrate **210**. The second electrode **320** may include at least one of a metal, a metal silicide or a metal compound. For example, the second electrode **330** may include aluminum.

(18) The first protecting insulation layer **340** may function as to protect the second electrode **330**. Further, the first protecting insulation layer **340** may function as to prevent the oxidation of the second electrode **330** caused by the moisture infiltration. The first protecting insulation layer **340** having a uniform thickness may be formed on the second electrode **330** in the via hole **310** along the surface of the via hole **310**. The first protecting insulation layer **340** having the uniform thickness on the second electrode **330** may be formed over the bottom surface and the side surface of the via hole **310**. Thus, the first protecting insulation layer **340** may have a cylindrical shape. Further, in order to effectively prevent the oxidation of the second electrode **330** caused by the moisture infiltration, the first protecting insulation layer **340** may include both ends protruded from

the both ends of the second electrode **330**. Therefore, the both ends of the first protecting insulation layer **340** may be protruded from the both ends of the second electrode **330** to effectively block a path of the moisture infiltration, thereby effectively preventing the oxidation of the second electrode **330**.

(19) The both ends of the first protecting insulation layer **340** may be protruded from the both ends of the second electrode **330** so that the both ends of the second electrode **330** may have the flat surface **S2** parallel to the surface **S1** of the upper substrate **210**. This may be illustrated in detail with reference to FIGS. 2C and 2D.

(20) Further, in order to prevent a galvanic corrosion caused by a difference between work functions of the second electrode **330** and the first protecting insulation layer **340**, the first protecting insulation layer **340** may include a nonmetal-based insulation material. When a protecting layer including titanium nitride may be used for protecting the second electrode **330** in the TSV structure **300**, a high difference between the work functions of titanium and aluminum may exist. Thus, the protecting layer including the titanium nitride may cause the galvanic corrosion as well as the oxidation caused by the moisture infiltration. However, the first protecting insulation layer **340** including the non-metal insulation layer may protect the second electrode **330** to prevent the galvanic corrosion.

(21) The first protecting insulation layer **340** may include oxide, nitride, or oxynitride. The first protecting insulation layer **340** may function as an anti-reflective layer for preventing a misalignment between processes. Thus, the nonmetal-based insulation material of the first protecting insulation layer **340** may have a reflectivity of about 30%, particularly, about 25% to about 35% in a photolithograph process using an I-line (365 nm). For example, the first protecting insulation layer **340** may include silicon oxynitride (SiON). When the first protecting insulation layer **340** may include the silicon oxynitride, the first protecting insulation layer **340** may have a thickness of about 330 Å to about 400 Å, or 1060 Å to about 1470 Å so as to meet the above-mentioned reflectivity.

(22) The second protecting insulation layer **350** may seal the second electrode **330** together with the first protecting insulation layer **340** to prevent the oxidation and corrosion caused by the moisture infiltration. Thus, the second protecting insulation layer **350** may be configured to cover the both ends having the flat surface **S2** of the second electrode **330**. The second protecting insulation layer **350** may include a nonmetal insulation material. In order to easily form the second protecting insulation layer **350** configured to cover the both ends having the flat surface **S2** of the second electrode **330** by the first protecting insulation layer **340**, the second protecting insulation layer **350** may include a material having low step coverage. For example, the second protecting insulation layer **350** may include silicon nitride (SiN).

(23) The pad structure **400** in the pad area **PA** may include a trench **410**, a first pad electrode **420**, a second pad electrode **430**, a capping layer **560**, a pad protection layer **440** and a pad opening **450**. The trench **410** may be formed at the stack structure **20**. The first pad electrode **420** may be formed on a surface of the trench **410**. The second pad electrode **430** may be formed on the first pad electrode **420**. The capping layer **560** may be formed on the stack structure **20** to fill the trench **410**. The pad protection layer **440** may be formed between the capping layer **560** and the second pad electrode **430**. The pad opening **450** may be formed at the capping layer **560** to expose the second pad electrode **430**. The capping layer **560** may include an insulation material.

(24) The trench **410** may function as to prevent a generation of a step at a stacked surface of the pad structure **400**, i.e., the surface **S1** of the upper substrate **210**. The trench **410** may be formed by partially etching the upper substrate **210**.

(25) The first pad electrode **420** having a uniform thickness may be formed along the surface of the trench **410**. That is, the first pad electrode **420** having the uniform thickness may be formed on a bottom surface and a side surface of the trench **410**. The first pad electrode **420** may include both ends extended to the upper substrate **210**. The other end of the first pad electrode **420** may be

extended to the TEA. The first pad electrode **420** may include a metal, a metal silicide or a metal compound. The first pad electrode **420** may be formed together with the first electrode **320** so that the first pad electrode **420** may include a material substantially the same as that of the first electrode **320**. For example, the first pad electrode **420** may include tungsten.

(26) The second pad electrode **430** having a uniform thickness may be formed on the first pad electrode **420**. That is, the second pad electrode **430** may have a plate shape having the uniform thickness over the bottom surface of the trench **410**. The second pad electrode **430** may include a metal, a metal silicide or a metal compound. The second pad electrode **430** may be formed together with the second electrode **330** so that the second pad electrode **430** may include a material substantially the same as that of the second electrode **330**. For example, the second pad electrode **430** may include aluminum.

(27) The pad protection layer **440** may protect the second pad electrode **430** between the processes. The pad protection layer **440** may be formed together with the first protecting insulation layer **340** so that the pad protecting layer **440** may include a material substantially the same as that of the first protecting insulation layer **340**. The pad protecting layer **440** may include oxide, nitride or oxynitride. For example, the pad protection layer **440** may include silicon oxynitride (SiON).

(28) According to example embodiments, the both ends of the second electrode **330** may be lower than the surface of the stack structure **20**, i.e., the surface **S1** of the upper substrate **210**. Further, the both ends of the first protecting insulation layer **340** may be upwardly protruded from the both ends of the second electrode **330**. Thus, the oxidation of the second electrode **330** caused by the moisture infiltration may be prevented.

(29) Because the both ends of the first protecting insulation layer **340** may be protruded from the both ends of the second electrode **330**, the both ends of the second electrode **330** may have the flat surface **S2** parallel to the surface **S1** of the upper substrate **210**. Thus, the second protecting insulation layer **350** configured to cover the second electrode **330** may be readily formed together with the first protecting insulation layer **340**. As a result, the oxidation of the second electrode **330** caused by the moisture infiltration may be effectively prevented.

(30) Furthermore, the first protecting insulation layer **340** and the second protecting insulation layer **350** may include the non-metal based insulation layer. Thus, although the moisture may infiltrate into the second electrode **330**, the galvanic corrosion may not be generated between the second electrode **330** and the first protecting insulation layer **340**, and between the second electrode **330** and the second protecting insulation layer **350**.

(31) As a result, the oxidation and the corrosion of the second electrode **330** caused by the moisture infiltration may be effectively prevented so that the semiconductor device including the TSV structure **300** may have improved reliability.

(32) FIGS. 2A to 2E are cross-sectional views illustrating an example of a method of fabricating a semiconductor device in accordance with example embodiments.

(33) Referring to FIG. 2A, the stack structure **20** including the lower structure **100** and the upper structure **200** may be prepared. The stack structure **20** may include the TEA and the PA. The TEA may correspond to an area where the TSV structure **300** in FIG. 1A may be to be formed. The PA may correspond to an area where the pad structure **400** in FIG. 1A may be to be formed.

(34) The lower structure **100** in the stack structure **20** may include the lower substrate **110** and the lower circuit **140**. The lower circuit **140** may be arranged in the lower insulating interlayer **150** on the lower substrate **110**. The lower circuit **140** may include the lower wiring **120** and the lower via pad **130** in the TEA. The upper structure **200** may include the upper substrate **210** and the upper circuit **240**. The upper circuit **240** may be arranged in the upper insulating interlayer **250** on the upper substrate **210**. The upper circuit **240** may include the upper wiring **220** and the upper via pad **230** in the TEA. The stack structure may be formed by bonding the lower structure **100** to the upper structure **200** while the upper insulating interlayer **250** and the lower insulating interlayer **150** face or contact to each other.

(35) The lower substrate **110** and the upper substrate **210** may include at least one of a bulk single crystalline silicon wafer, a silicon-on-insulator (SOI) wafer, a compound semiconductor wafer such as Si—Ge, a wafer including a silicon epitaxial layer, or others. The lower wiring **120**, the lower via pad **130**, the upper wiring **220** and the upper via pad **230** may include a metal such as tungsten, aluminum, copper, etc., a metal silicide such as tungsten silicide, titanium silicide, etc., a metal compound such as tungsten nitride, titanium nitride, etc., a doped silicon, or others.

(36) The via hole **310** may be formed at the TEA. The via hole **310** may be formed by, for example, etching the upper substrate **210**, the upper insulating interlayer **250** and the lower insulation interlayer **150** using a mask pattern on the upper substrate **210** as an etch mask until the lower via pad **130** may be exposed. The via hole **310** may be configured to expose the upper via pad **230**. Alternatively, the via hole **310** may be formed through the upper via pad **230**.

(37) The trench **410** may be formed at the pad area PA. The trench **410** may provide a space for the pad structure **400** as shown in in FIG. 1A. Further, the trench **410** may prevent the generation of the step at the surface of the stack structure **20** caused by the pad structure **400**. The upper substrate **210** may be partially etched to form the trench **410**. In some implementations, the etching process for forming the via hole **310** and the trench **410** may include an anisotropic dry etch process.

(38) Referring to FIG. 2B, a first conductive layer **510** having a uniform thickness may be formed on the surface of the stack structure with the via hole **310** and the trench **410**. The first conductive layer **510** may have a thickness that may not fully fill the via hole **310** and the trench **410**. The first conductive layer **510** may include a metal, a metal silicide or a metal compound. For example, the first conductive layer **510** may include tungsten.

(39) Although not depicted in drawings, the first conductive layer **510** may be selectively etched to form the first electrode **320** in the TEA and the first pad electrode **420** in the PA. The first electrode **320** may be formed along the surface of the via hole **310**. The first electrode **320** may be partially extended to the upper surface of the upper substrate **210**. Thus, the first electrode **320** may be configured to cover the bottom surface and the side surface of the via hole **310** and to partially cover the upper substrate **210** adjacent to the via hole **310**. The first pad electrode **420** may be formed along the surface of the trench **410**. The first pad electrode **420** may be partially extended to the upper surface of the upper substrate **210**. Thus, the first pad electrode **420** may be configured to cover the bottom surface and the side surface of the trench **410** and to partially cover the upper substrate **210** adjacent to the trench **410**. The first electrode **320** and the first pad electrode **420** that are extended to the upper surface of the upper substrate **210** may be electrically connected with each other.

(40) A second conductive layer **520** may be formed on the surface of the stack structure **20** with the via hole **310**, the first electrode **320**, the trench **410** and the first pad electrode **420**. The second conductive layer **520** may have a thickness that may not fully fill the via hole **310** and the trench **410**. The second conductive layer **520** may have a uniform thickness along the surface of the stack structure **20**. Alternatively, a portion of the second conductive layer **520** on the bottom surfaces of the via hole **310** and the trench **410** may have a thickness greater than a thickness of a portion of the second conductive layer **520** on the side surfaces of the via hole **310** and the trench **410**. To configure the second conductive layer **520** to have a greater thickness on bottom surfaces of the via hole **310** and the trench **410**, a deposition condition can be adjusted by, for example, controlling a pressure such that a vertical deposition rate is greater than a horizontal deposition rate. The second conductive layer **520** may include at least one of a metal, a metal silicide, or a metal compound. For example, the second conductive layer **520** may include aluminum.

(41) An insulation layer **530** having a uniform thickness may be formed on the second conductive layer **520** along the surface of the stack structure **20** with the via hole **310** and the trench **410**. The insulation layer **530** may have a thickness that may not fully fill the via hole **310** and the trench **410**. The insulation layer **530** may include a material having an etching selectivity with respect to a material of the second conductive layer **520**. Particularly, the insulation layer **530** may include an

insulation material having an etching selectivity with respect to an etchant for etching the second conductive layer **520**. The insulation layer **530** may include at least one of oxide, nitride, or oxynitride. For example, the insulation layer **530** may include a nonmetal-based insulation material.

(42) The insulation layer **530** may act as the anti-reflective layer in forming a pad mask **540** for a second pad electrode in FIG. 2C. In some implementations, in a photolithograph process using an I-line (365 nm), the insulation layer **530** may have a material with a reflectivity in a range about 25% to about 35%, for example, about 30%.

(43) Thus, the insulation layer **530** may include silicon oxynitride (SiON). When the insulation layer **530** may include the silicon oxynitride, the first protecting insulation layer **340** may have a thickness of about 330 Å to about 400 Å, or 1060 Å to about 1470 Å so as to meet the above-mentioned reflectivity.

(44) Referring to FIG. 2C, the pad mask **540** may be formed on the insulation layer **530** in the PA. The insulation layer **530** and the second conductive layer **520** may be etched using the pad mask **540** as an etch barrier until the first conductive layer **510** may be exposed. The etch process may be performed with the TEA being exposed. The etch process may include an anisotropic dry etch process. For example, an etch-back process may be performed in the TEA.

(45) Therefore, the second pad electrode **430** and the pad protection layer **440** may be formed in the PA. Further, the second electrode **330** and the first protecting insulation layer **340** may be formed in the TEA. Because the etch process may be performed with the TEA being fully exposed, the second electrode **330** and the first protecting insulation layer **340** may have cylindrical shapes. The both ends of the second electrode **330** may be positioned below the surface of the upper substrate **210**. In some implementations, the first protecting insulation layer **340** may include the material having a etching selectivity with respect to the material of the second conductive layer **520**, for example, SiON. Thus, the both ends of the first protecting insulation layer **340** may be upwardly protruded from the both ends of the second electrode **330**. Thus, the both ends of the second electrode **330** may have the flat surface S2 in FIG. 1B parallel to the surface of the upper substrate **210**.

(46) In contrast, when the first protecting insulation layer **340** may not have the etch selectivity with respect to the second conductive layer **520**, the both ends of the first protecting insulation layer **340** may not be upwardly protruded from the both ends of the second electrode **330** so that the first protecting insulation layer **340** without the upward protrusion may not effectively block the path of the moisture infiltration. Further, in this case, the both ends of the second electrode **330** may have sharp ends so that the second protecting insulation layer **350** may not fully cover the both ends of the second electrode **330** by following processes.

(47) The pad mask **540** may then be removed by an ashing process and/or a stripping process.

(48) Referring to FIG. 2D, a via mask **550** may be formed on the stack structure **20**. The via mask may have an opening configured to expose the via hole **310**. The second protecting insulation layer **350** may be formed on the surfaces S2 of the both ends of the second electrode **330**. In order to selectively form the second protecting insulation layer **350** on the surfaces of the both ends of the second electrode **330**, the second protecting insulation layer **350** may be formed under deposition conditions including low step coverage characteristics. For example, the second protecting insulation layer **350** may be formed by setting a vertical deposition rate greater than a horizontal deposition rate. For example, the vertical deposition rate can be set to be greater than that of the horizontal deposition rate by controlling the pressure.

(49) The second protecting insulation layer **350** may include at least one of oxide, nitride, or oxynitride. For example, the second protecting insulation layer **350** may include a non-metal-based insulation material. In order to selectively form the second protecting insulation layer **350** on the both ends of the second electrode **330**, the second protecting insulation layer **350** may include an insulation material having low step coverage characteristics. For example, the second protecting

insulation layer **350** may include silicon nitride.

(50) Because the both ends of the second electrode **330** may have the flat surface **S2** in FIG. **1B** parallel to the surface **S1** of the upper substrate **210** in FIG. **1B**, the second protecting insulation layer **350** can fully cover the both ends of the second electrode **330**. Thus, the second electrode **330** may be fully sealed by the first protecting insulation layer **340** and the second protecting insulation layer **350**. As a result, the oxidation of the second electrode **330** caused by the moisture infiltration may be effectively prevented.

(51) The via mask **550** may then be removed by an ashing process and/or a stripping process. By removing the via mask **550**, the second protecting insulation layer **350** on an unnecessary region may also be removed.

(52) Referring to FIG. **2E**, the capping layer **560** may be formed on the surface of the stack structure **20**. The capping layer **560** may include an insulation material.

(53) The capping layer **560** and the pad protection layer **440** may be selectively etched to form the pad opening **450** configured to expose the second pad electrode **430**.

(54) Therefore, the TSV structure **300** may be completed. The TSV structure **300** may include the via hole **310**, the first electrode **320**, the second electrode **330**, the first protecting insulation layer **340** and the second protecting insulation layer **350**. The via hole **310** may be formed at the stack structure **20**. The first electrode **320** may be formed on the surface of the via hole **310**. The second electrode **330** may be formed on the first electrode **320** along the surface of the via hole **310**. The first protecting insulation layer **340** may be formed on the second electrode **330** along the surface of the via hole **310**. The second protecting insulation layer **350** may be configured to cover the both ends of the second electrode **330**.

(55) Further, the pad structure **400** may also be completed. The pad structure **400** may include the trench **410**, the first pad electrode **420**, the second pad electrode **430**, the capping layer **560**, the pad protecting layer **440** and the pad opening **450**. The trench **410** may be formed at the stack structure **20**. The first pad electrode **420** may be formed on the surface of the trench **410**. The second pad electrode **430** may be formed on the first pad electrode **420**. The capping layer **560** may be configured to cover the stack structure **20** and to fill the trench **410**. The pad protection layer **440** may be formed between the capping layer **560** and the second pad electrode **430**. The pad opening **450** may be formed at the capping layer **560** to expose the second pad electrode **430**.

(56) While this patent document contains many specifics, these should not be construed as limitations on the scope of any invention or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular inventions. Certain features that are described in this patent document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(57) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Moreover, the separation of various system components in the embodiments described in this patent document should not be understood as requiring such separation in all embodiments.

(58) Only a few implementations and examples are described and other implementations, enhancements and variations can be made based on what is described and illustrated in this patent document.

Claims

1. A semiconductor device, comprising: a substrate including a via hole formed in the substrate; a first electrode disposed along an inner surface of the via hole; and a second electrode disposed along on a surface of the first electrode to have a height lower than a height of the first electrode, wherein the second electrode is directly connected to the first electrode.
2. The semiconductor device of claim 1, further comprising: a protecting layer disposed to cover an exposed surface of the second electrode which is not in contact with the first electrode.
3. The semiconductor device of claim 2, wherein the protecting layer comprises a first protecting insulation layer disposed on a sidewall of the second electrode positioned at a sidewall of the via hole.
4. The semiconductor device of claim 3, wherein a height of the first protecting insulation layer is greater than the height of the second electrode.
5. The semiconductor device of claim 3, wherein the first protecting insulation layer comprises a material having a reflectivity of about 25% to about 35% in an I-line photolithography process.
6. The semiconductor device of claim 3, wherein the protecting layer further comprises a second protecting insulation layer disposed on an upper surface of the second electrode positioned between the first electrode and the first protecting insulation layer.
7. The semiconductor device of claim 6, wherein an upper surface of the first protecting insulation layer is closer to the upper surface of the substrate than an upper surface of the second protecting insulation layer.
8. The semiconductor device of claim 6, wherein at least one of the first protecting insulation layer and the second protecting insulation layer comprises a non-metal insulation material.
9. The semiconductor device of claim 6, wherein at least one of the first protecting insulation layer and the second protecting insulation layer comprises at least one of an oxide material, a nitride material, or an oxynitride material.
10. The semiconductor device of claim 1, further comprising a capping layer configured to cover the substrate and fill the via hole.
11. The semiconductor device of claim 1, wherein the first electrode further includes an extending portion that extends to an upper surface of the substrate outside the via hole.
12. The semiconductor device of claim 11, further comprising a pad structure disposed on the upper surface of the substrate, wherein the pad structure is connected to the via hole by the extending portion of the first electrode.
13. A semiconductor device, comprising: a substrate including a via hole formed in the substrate; a first electrode disposed along an inner surface of the via hole; and a second electrode disposed along on a surface of the first electrode to have a height lower than a height of the first electrode; a protecting layer disposed to cover an exposed surface of the second electrode which is not in contact with the first electrode; wherein the second electrode is electrically connected to the first electrode, wherein the protecting layer comprises a first protecting insulation layer disposed on a sidewall of the second electrode positioned at a sidewall of the via hole, and a second protecting insulation layer disposed on an upper surface of the second electrode positioned between the first electrode and the first protecting insulation layer.
14. The semiconductor device of claim 13, wherein an upper surface of the first protecting insulation layer is closer to the upper surface of the substrate than an upper surface of the second protecting insulation layer.
15. The semiconductor device of claim 13, wherein at least one of the first protecting insulation layer and the second protecting insulation layer comprises a non-metal insulation material.
16. The semiconductor device of claim 13, wherein at least one of the first protecting insulation layer and the second protecting insulation layer comprises at least one of an oxide material, a

nitride material, or an oxynitride material.

17. A semiconductor device, comprising: a substrate including a via hole disposed in the substrate; a first electrode disposed along an inner surface of the via hole and extending to an upper surface of the substrate; a second electrode continuously disposed along a surface of the first electrode, the second electrode being in direct contact with the first electrode; and a protecting layer disposed to cover an exposed surface of the second electrode, wherein a height of the second electrode is lower than a height of the first electrode.

18. The semiconductor device of claim 17, wherein the protecting layer comprises: a first protecting insulation layer disposed on a sidewall of the second electrode positioned at a sidewall of the via hole; and a second protecting insulation layer disposed on an upper surface of the second electrode positioned between the first electrode and the first protecting insulation layer.

19. The semiconductor device of claim 18, wherein at least one of the first protecting insulation layer and the second protecting insulation layer comprises a non-metal insulation material.
